LOST AND FOUND

AI-POWERED ITEM RECOVERY

Never lose a trace of your belongings.

LostLink is the first intelligent lost and found system for SRM University. Our AI matching engine works 24/7 to reconnect students with their items.

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ACCOUNT ACCESS

Smart Campus Lost & Found System

PROBLEM STATEMENT

The Challenge on Campus

- High Volume of Lost Items: Students frequently lose IDs, electronics, and books across a large campus.
- Inefficient Processes: Traditional WhatsApp groups or physical lost-and-found boxes are unorganized and rely heavily on manual searching.
- Fraudulent Claims: Anyone can claim a found item if there is no secure verification process in place.
- Delayed Returns: Lack of real-time matching leads to items remaining unclaimed for long periods.

PROPOSED SOLUTION

01

- Automated Matching: Uses Natural Language Processing (NLP) to connect "Lost" and "Found" reports automatically.

02

- Secure Authentication: Ensures items are only returned to their rightful owners via a robust challenge-response verification system.

03

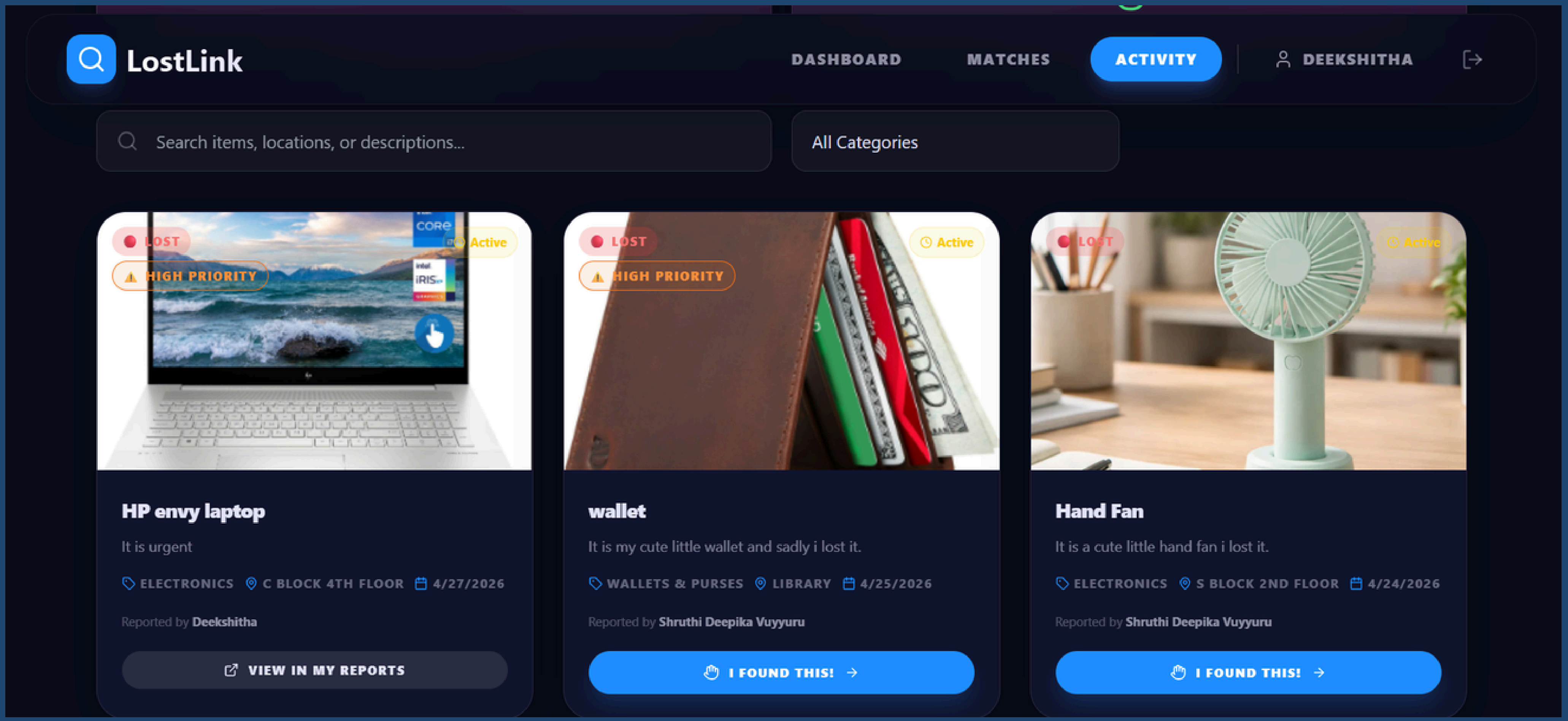
- Community Driven: Rewards users with "Reputation Scores" for honestly reporting found items.

KEY FEATURES

1. **Role-Dynamic Dashboard:** Tailored views whether you are looking for a lost item or reporting a found one.
2. **Live Activity Feed:** A real-time, searchable feed of all active lost items on campus.
3. **"I Found This" Workflow:** One-click process for finders to notify losers directly from the activity feed.
4. **AI Match Results:** A dedicated page showing potential matches ranked by confidence percentage.
5. **Secure Ownership Challenge:** Custom security questions set by the finder that the claimant must answer correctly.

AI ARCHITECTURE & IMPLEMENTATION

- INTELLIGENT PRIORITY AGENT
- FORWARD CHAINING EXPERT SYSTEM
- CROSS-LANGUAGE MICROSERVICE
- DATA REPRESENTATION & UI



LOGIC:

```
# "KNOWLEDGE BASE" OF RULES
HIGH_VALUE_KEYWORDS = ['WALLET', 'PURSE', 'CASH', 'MONEY', 'LAPTOP']
CRITICAL_DOC_KEYWORDS = ['PASSPORT', 'ID', 'VISA', 'CERTIFICATE']

# FORWARD CHAINING LOGIC
IF ANY(WORD IN TEXT FOR WORD IN CRITICAL_DOC_KEYWORDS):
   URGENCY_LEVEL = "CRITICAL"
   REASON = "CONTAINS CRITICAL OFFICIAL DOCUMENT KEYWORDS."
```

TECHNOLOGY STACK

- Frontend: React.js, Tailwind CSS, Framer Motion (for smooth micro-animations and a premium UI).
- Backend: Node.js & Express.js (RESTful API architecture).
- Database: MongoDB & Mongoose (Flexible schema for items and claims).
- AI/NLP: natural library (Node.js) for text processing and algorithmic matching.
- Security: JSON Web Tokens (JWT) for session management and Bcrypt for password hashing.

SYSTEM ARCHITECTURE & WORKFLOW

1. Report: User A reports an item as "Lost". User B reports a similar item as "Found".
2. Analyze: The backend algorithm constantly scans active reports.
3. Match & Notify: User A is notified of a high-confidence match.
4. Claim: User A initiates a claim on User B's found item.
5. Verify: User A correctly answers User B's security questions.
6. Resolve: Item status changes to "Recovered", and User B earns a reputation point.

CORE LOGIC – THE AI MATCHING ALGORITHM

- Text Similarity (TF-IDF & Cosine Similarity): Analyzes the item descriptions and locations to find semantic overlaps (e.g., "Airpods" and "White Apple Earbuds").
- Fuzzy Title Matching (Levenshtein Distance): Calculates the edit distance between titles to handle typos and spelling mistakes.
- Time Decay Factor: Matches reported closer in time receive a higher confidence score, calculated using an exponential decay mathematical function.
- Scoring Weightage:
  - 35% Description similarity
  - 30% Title fuzzy match
  - 20% Exact category match
  - 10% Location proximity/similarity
  - 5% Time relevance

CORE LOGIC – SECURE CLAIM VERIFICATION

- Challenge-Response System: When reporting a Found item, the finder sets a mandatory security question (e.g., "What is the wallpaper on the phone?").
- Fuzzy Answer Evaluation: The backend doesn't just do strict exact matches. The algorithm approves claims if:
  - Exact match (case-insensitive)
  - Partial inclusion (e.g., user types "butterfly" for "butterfly company")
  - Significant keyword overlap
- Automated Status Updates: Upon passing verification, both the Lost and Found database records are automatically locked and marked as "Matched/Recovered".

IMPACT & FUTURE SCOPE

Current Impact

- Reduces time taken to recover lost belongings, builds trust within the campus, and cleans up spam on student messaging groups.

Future Scope

- Image Recognition: Integrating computer vision AI to match items based on uploaded photos, reducing reliance on text descriptions.
- Push Notifications: Adding progressive web app (PWA) features for native mobile push alerts when a match is found.
- Admin Panel: Creating a dashboard for campus security to monitor and override reports if necessary.

CONCLUSION

1. PROJECT SUMMARY

GOAL ACHIEVED: SUCCESSFULLY TRANSFORMED A STANDARD CAMPUS APPLICATION INTO AN AI-DRIVEN PLATFORM (LOSTLINK).

2. REAL-WORLD IMPACT

ULTIMATELY, THIS PROJECT SERVES AS A MODEL FOR 'AI FOR SOCIAL GOOD,' DEMONSTRATING HOW INTELLIGENT AUTOMATION CAN BE LEVERAGED NOT JUST FOR CORPORATE EFFICIENCY, BUT TO BUILD MORE EMPATHETIC, COOPERATIVE, AND TIGHTLY-KNIT COMMUNITIES.

"IN CONCLUSION, WE DIDN'T JUST BUILD A CRUD APP; WE BUILT AN INTELLIGENT AGENT THAT ACTIVELY SOLVES A REAL CAMPUS PROBLEM. BY APPLYING THE FORWARD CHAINING RULES WE LEARNED IN CLASS, THE SYSTEM NOW WORKS FOR THE USERS. MOVING FORWARD, WE SEE HUGE POTENTIAL TO EXPAND THIS WITH COMPUTER VISION AND PREDICTIVE MODELS."

THANK YOU

GITHUB REPO